

DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Thermodynamics & Heat Transfer | Course Code: 21ME32 | Credits: 4

Exam Session: End Semester Exam 2024

Max Marks: 80 | Duration: 3 Hours

INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

- Q1.** A steam power plant operates on an ideal Rankine cycle between boiler pressure 3 MPa and condenser pressure 10 kPa. If steam leaves boiler as superheated steam at 350°C, calculate: (i) Turbine work output, (ii) Pump work input, (iii) Heat added in boiler, and (iv) Thermal efficiency of the cycle.

[8 Marks]